

12_FINAL_REVIEW.md

TASK 12

FINAL ENGINEERING REVIEW & RELEASE

Status

Mandatory

Owner

Principal Engineer

Supporting Guides

All Engineering Guides

Estimated Time

1 to 2 Days

Dependencies

00_PROJECT_CONTEXT.md

00_ENGINEERING_RULES.md

00_PROJECT_MEMORY.md

12_PRINCIPAL_ENGINEER.md

01_PROJECT_SETUP.md

02_REQUIREMENTS.md

03_ARCHITECTURE.md

04_DATABASE.md

05_UI_DESIGN.md

06_BACKEND.md

07_FRONTEND.md

08_SECURITY.md

09_TESTING.md

10_FORMAL_METHODS.md

11_DOCUMENTATION.md

Mission

Perform a complete engineering review of the Secure Dance Academy Management System.

Review every requirement.

Review every module.

Review every security control.

Review every document.

Review every test.

Review every diagram.

Approve the project only when it satisfies professional engineering standards and distinction-level coursework expectations.

Primary Objectives

Validate the entire system.

Ensure implementation matches requirements.

Confirm security controls.

Review architecture.

Review maintainability.

Review scalability.

Review documentation.

Review presentation readiness.

Prepare the final submission.

Engineering Philosophy

Assume nothing.

Verify everything.

Evidence is more valuable than assumptions.

Professional software is measured by quality, not quantity.

Every review should improve the project.

Phase 1

Requirements Verification

Verify

All functional requirements implemented.

All non-functional requirements satisfied.

Business rules enforced.

Security requirements implemented.

Privacy requirements implemented.

Legal requirements addressed.

Acceptance criteria satisfied.

Requirements traceability complete.

Nothing should remain unimplemented.

Phase 2

Architecture Review

Verify

Clean Architecture maintained.

Feature-based architecture maintained.

Folder structure consistent.

No circular dependencies.

Business logic isolated.

Database access isolated.

Security layers maintained.

Architecture documentation updated.

Every module has one responsibility.

Phase 3

Database Review

Verify

Normalization.

Relationships.

Indexes.

Constraints.

Transactions.

Soft Deletes.

Audit Logs.

Migration Success.

Seed Data.

Performance.

Backup Strategy.

RLS Policies.

Database should support future growth.

Phase 4

Backend Review

Verify

Authentication.

Authorization.

Validation.

Business Logic.

Repositories.

Transactions.

Audit Logging.

Error Handling.

API Documentation.

Performance.

No duplicated logic.

No unnecessary complexity.

Phase 5

Frontend Review

Verify

Responsive Layouts.

Navigation.

Authentication.

CRUD Pages.

Forms.

Tables.

Charts.

Accessibility.

Performance.

Loading States.

Empty States.

Error States.

Design Consistency.

The application should feel like enterprise software.

Phase 6

Security Review

Verify

OWASP Top 10.

OWASP ASVS.

NIST SSDF.

Authentication.

Authorization.

Session Management.

Validation.

Rate Limiting.

Secure Headers.

Secrets Management.

Audit Logging.

Residual Risks.

No Critical vulnerabilities should remain.

Phase 7

Testing Review

Verify

Unit Tests.

Integration Tests.

API Tests.

Playwright Tests.

OWASP ZAP.

SonarQube.

Performance Tests.

Accessibility Tests.

Regression Tests.

Coverage Targets.

Review every report.

Phase 8

Formal Methods Review

Verify

Finite State Machines.

Petri Nets.

Transition Tables.

Reachability.

Deadlock Analysis.

Safety.

Liveness.

Traceability.

Formal verification should match implementation.

Phase 9

Documentation Review

Review

SRS.

Architecture.

Database.

API.

Security.

Testing.

Deployment.

Developer Guide.

User Guide.

Administrator Guide.

README.

Final Report.

Harvard References.

Documentation should accurately represent the application.

Phase 10

Performance Review

Measure

Application Startup.

Dashboard Loading.

Search Speed.

API Response Time.

Database Queries.

Memory Usage.

Bundle Size.

Rendering.

Performance should remain acceptable.

Phase 11

Accessibility Review

Verify

WCAG 2.2 AA.

Keyboard Navigation.

Screen Readers.

ARIA Labels.

Contrast.

Responsive Layouts.

Semantic HTML.

Accessibility should be complete.

Phase 12

GitHub Review

Verify repository quality.

README.

Folder Structure.

Documentation.

Commit History.

Screenshots.

License.

Environment Template.

Setup Instructions.

Repository should appear production ready.

Phase 13

Coursework Review

Verify every marking area.

Design

Architecture

Diagrams

Requirements

Development

Frontend

Backend

Database

Security

Security Controls

Testing

Evidence

Reports

Formal Methods

FSM

Petri Nets

Verification

Documentation

Presentation

No assessment criterion should lack supporting evidence.

Deliverables

Produce

Engineering Review Report.

Architecture Review.

Security Review.

Database Review.

Backend Review.

Frontend Review.

Testing Review.

Performance Review.

Accessibility Review.

Formal Methods Review.

Documentation Review.

Technical Debt Register.

Improvement Roadmap.

Release Notes.

Submission Checklist.

Presentation Checklist.

Final Approval Report.

Technical Debt Review

Identify

Code Duplication.

Large Components.

Large Functions.

Architecture Violations.

Performance Issues.

Security Weaknesses.

Documentation Gaps.

Testing Gaps.

Record

Priority.

Impact.

Owner.

Recommended Solution.

Release Criteria

Approve only if

No Critical Security Issues.

No High Severity Bugs.

Tests Pass.

Architecture Approved.

Documentation Complete.

Formal Methods Complete.

Performance Acceptable.

Accessibility Passes.

Deployment Successful.

Project Demonstration Ready.

Final Submission Checklist

Project builds successfully.

Application deploys successfully.

Authentication works.

Authorization works.

Every module works.

Database migrations succeed.

Documentation complete.

Diagrams complete.

Screenshots prepared.

Presentation prepared.

GitHub repository clean.

README complete.

Environment template included.

Update Project Memory

Mark

Final Review

Completed

Record

Release Version.

Review Date.

Outstanding Improvements.

Technical Debt.

Final Risks.

Submission Status.

Project Status

Completed.

Final Engineering Questions

Before approving the release, answer every question.

Does the implementation satisfy every functional requirement?

Does the architecture follow Clean Architecture?

Does every security control have evidence?

Does every API follow the approved standards?

Does every database entity have a purpose?

Does the frontend match the approved designs?

Does the testing provide measurable confidence?

Do the Formal Methods accurately represent implementation?

Is the documentation complete and professional?

Is the GitHub repository portfolio quality?

Would an experienced software engineer approve this project?

Would this system be suitable for deployment in a real organization?

If any answer is "No", identify the issue, correct it, and repeat the review.

Definition of Done

The project is complete only when every engineering discipline agrees that the software is production ready.

The final submission should demonstrate secure software engineering, modern architecture, professional implementation, comprehensive testing, formal verification and high-quality documentation.

The Secure Dance Academy Management System should be ready for:

- MSc distinction-level assessment
- Professional GitHub portfolio
- Technical interviews
- Live deployment
- Future expansion

Release approval is granted only after every checklist has been satisfied and every major engineering decision has been justified with evidence.